CONCEPT MAP

Classification of Elements

- MODERN PERIODIC TABLE
  - By Mosley-(1912) The physical and chemical properties of elements are periodic functions of their atomic number
  - Periods and Groups
    - Groups:
      - Vertical columns contain elements having similar outer electronic configurations.
      - (18 Groups)
    - Periods:
      - Horizontal rows. Period number corresponds to the highest principal quantum number (n) of the elements.
      - Periods 1 to 6 contains 2,8,8,18,18,32 elements respectively.
      - Period 6 contains 14 elements (lanthanides)
      - Period 7 contains 14 elements (actinides)
  - Blocks
    - s- block:
      - group-1 : hydrogen and alkali metals
      - group 2 : alkaline earth metals
    - P block:
      - Group 13
      - Group 14
      - Group 15 :Pnictogens
      - Group-16 :Chalcogens
      - Group-17 : Halogens
      - Group 18: Noble gases
    - d block:
      - Contains 4 series
    - F block:
      - Contains 2 series
      - I series : lanthanides
      - II Series : actinides
- PERIODIC TRENDS IN PROPERTIES OF ELEMENTS
  - TRENDS IN PHYSICAL PROPERTIES
    - Atomic Radii & Ionic Radii
      - Along period decreases
      - Within a group increases
    - Ionisation Enthalpy,
      - Along period decreases
      - Within a group increases
    - Electron affinity,
    - Electronegativity
  - TRENDS IN CHEMICAL PROPERTIES
    - Valence or oxidation state
    - Strength of oxyacids of non-metal
    - Acidity of hydrides
    - Acidity of oxides
    - Anomalous behavior of elements of II period: diagonal relationship : some elements of II period show similarities with diagonally placed III period elements
      - Eg : Li & Mg , Be & Al

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